جداول الفيزياء العامة: الأنظمة والوحدات والأبعاد

General Physics: Systems of Units, Conversion Factors, and Dimensional Analysis

Table (1) Units of the fundamental quantities in different systems of units

Quantity	CGS	MKS (SI)	FPS
Length	Centimeter (cm)	Meter (m)	Foot (ft)
Mass	Gram (g)	Kilogram (kg)	Pound (p)
Time	Second (s)	Second (s)	Second (s)

Table (2) Units' conversion factors

Length	Mass	Time
1 cm = 10^{-2} m	1 g = 10^{-3} kg	1 min = 60 s
1 mm = 10^{-3} m	1 Ton = 10^{3} kg	1 h = 60 min = 3600 s
1 inch = 2.54 cm = 2.54 × 10^{-2} m	1 pound (lb) = 0.453 kg	1 day = 24 h = 86400 s
1 foot = 12 inch = 0.3048 m	1 ounce (oz) = 28.3495 g	1 y = 365.25 day
1 yard = 3 ft = 0.9144 m		
1 mile = 1760 yd = 1609.3 m = 1.6093 km		
1 Light year = 9.46 × 10^{15} m		

Table (3) Scientific Notation Prefixes

Power	Prefix	Symbol	Power	Prefix	Symbol
10^{-1}	deci	d	10^{3}	kilo	k
10^{-2}	centi	c	10^{6}	mega	M
10^{-3}	milli	m			

Table (4) Dimensions of fundamental physical quantities

Quantity	Dimension
Length	L
Mass	M
Time	T

Example 1.2 & Practice Exercise 1.2: Derived Physical Quantities

Physical Quantity	Math. definition	Dimensions	MKS Unit
Speed (v)	$v = x / t$	$L / T = LT^{-1}$	$m \cdot s^{-1}$
Acceleration (a)	$a = v / t$	$LT^{-1} / T = LT^{-2}$	$m \cdot s^{-2}$
Force (F)	$F = m\,a$	MLT^{-2}	$kg \cdot m \cdot s^{-2}$
Linear momentum (p)	$p = m\,v$	MLT^{-1}	$kg \cdot m \cdot s^{-1}$
Pressure (P)	$P = F / A$	$ML^{-1}T^{-2}$	$kg \cdot m^{-1} \cdot s^{-2}$
Work (W)	$W = F\,x$	ML^2T^{-2}	$kg \cdot m^2 \cdot s^{-2}$
Kinetic Energy (T)	$T = \frac{1}{2}\,m\,v^2$	ML^2T^{-2}	$kg \cdot m^2 \cdot s^{-2}$
Potential Energy (V)	$V = m\,g\,h$	ML^2T^{-2}	$kg \cdot m^2 \cdot s^{-2}$